Critical Review of Agbola and Acupan (2010): An Empirical Analysis of International Labour Migration in the Philippines

Lian Cataytay
University of Southeastern Philippines · October 22, 2026

Agbola and Acupan (2010) estimate how domestic conditions shape emigration from the Philippines with annual data for 1975–2005 and an error correction model (ECM). The time-series steps are careful, but a small sample, weak proxies, and missing pull factors limit the conclusions.

1 Research Problem / Gap

Labour export began in the 1970s as a stop-gap for unemployment and a balance-of-payments crisis but became structural: about 8.2 million Filipinos lived or worked abroad and sent home US$14.4 billion in 2007 (Agbola & Acupan, 2010). Policy is ambiguous: Republic Act No. 8042 states that the State "does not promote overseas employment" (Section 2c), yet that deployment "shall be encouraged" (Section 2i) (as cited in Agbola & Acupan, 2010, p. 388). The gap is that the determinants of emigration "are not well established" (p. 386). The research question is how economic, demographic, and political factors affect the size of emigration. ***Assessment.*** The question is clear and policy-relevant, but the gap is asserted rather than demonstrated: earlier Philippine studies are not reviewed.

2 Theoretical Framework

The authors draw on four strands (as cited in Agbola & Acupan, 2010): expected-utility models (Ravenstein, 1885; Todaro, 1969); human capital theory (Hicks, 1939; Sjaastad, 1962), which predicts that education raises emigration; the Roy effect (Borjas, 1987), in which inequality determines which skill groups leave; and the gravity model (Greenwood, 1975; Adams & Page, 2003), which links migration to income, population, and cost. Together they yield one macro push-factor equation for the migration share with nine domestic explanatory variables. ***Assessment.*** The framework is eclectic: theory mainly fixes the expected signs, and although the gravity model is bilateral, no destination-side (pull) variable enters.

3 Data

The study uses annual national aggregates for 1975–2005 ($T = 31$), all secondary. The migration share (temporary plus permanent migrants over population) comes from the Commission on Filipinos Overseas (CFO) and the National Statistical Coordination Board (NSCB). Macro series come from the IMF, BSP, and NSCB. Key proxies are airfare (Manila to the top 24 destinations) for migration cost, adult literacy for education, and leadership changes and uprisings for political instability. ***Assessment.*** The Gini source is not stated (the Philippine Gini is published only every few years, implying undisclosed interpolation); the migration share pools temporary and permanent migrants; airfare is a lower bound on cost; and near-universal literacy barely varies.

4 Methodology

The design is a five-step time-series procedure: (a) ADF and Phillips–Perron unit root tests treat the series as I(1); (b) Engle–Granger, Johansen, and Pesaran bounds tests confirm a long-run relationship; (c) a general-to-specific reduction yields a parsimonious model; (d) an unrestricted

ECM is estimated by OLS; and (e) all diagnostic tests are passed. ***Assessment.*** The sequence is textbook, but three problems remain. The Johansen test finds five to eight cointegrating vectors, yet only one error correction term (ECT) is kept. Per capita income fails the Phillips–Perron test even in first differences (−2.123 against a 10% critical value of −2.623). And 31 observations for 10 estimated terms leave little power, with no structural-break tests (e.g., 1986, 1997).

5 Key Findings

Table 1. Error correction model results

Explanatory variable	Coefficient	*t*-statistic	Result (5%)
Unemployment	+0.236	4.629	Significant (+)
Adult literacy	−0.659	−5.225	Significant (−)
Political instability	−0.482	−3.444	Significant (−)
Population density	−0.254	−2.756	Significant (−)
Inequality, inflation, cost, income	—	$\|t\| < 1.6$	Not significant

Notes: Adjusted $R^2 = 0.63$; $F = 5.848$ ($p = 0.000$). *Source:* Adapted from Agbola and Acupan (2010, p. 392).

The ECT (−0.776, $t = -4.224$) implies that about 78% of any deviation from equilibrium is corrected within a year. Unemployment raises emigration, consistent with emigration as a safety valve for surplus labour. Adult literacy lowers it, contradicting human capital theory; the authors attribute this to the low-skilled profile of migrants. Political instability and population density also lower emigration. The other variables are not significant, so the Roy effect is rejected. The authors recommend better migration statistics, bilateral protection agreements, and a domestic economy able to absorb labour. ***Assessment.*** The low-skill explanation conflicts with evidence the paper itself cites: Alburo and Abella (2002, as cited in Agbola & Acupan, 2010) found that professional outflows exceeded domestic additions. Insignificant results may also reflect low power.

6 Areas for Further Research

The authors acknowledge that migration statistics are hard to compile, that airfare is only the minimum cost of migration, that the density result needs further research, and that their data do not “permit us to move beyond speculation” on political instability (p. 394). They call for explaining the density effect and for better monitoring of migration statistics. I suggest four further directions: (a) disaggregate the migration share by type (temporary vs. permanent) and by skill; (b) add pull factors through a bilateral panel gravity model with destination wages and labour demand; (c) extend the sample past 2005 and test for structural breaks (1997, 2008, COVID-19); and (d) use household survey data to test human capital selection.

Overall Assessment

Agbola and Acupan (2010) offer useful early evidence that domestic push factors, led by unemployment, shape Filipino emigration. Given $T = 31$, weak proxies, and no pull factors, the results are a starting point: **push factors at home matter, but they are not the whole story.**

Reference

Agbola, F. W., & Acupan, A. B. (2010). An empirical analysis of international labour migration in the Philippines. *Economic Systems, 34*(4), 386–396. https://doi.org/10.1016/j.ecosys.2010.03.002